PyRAMIDS Manual

A Python package for Radiation, Magnetoelectric Interactions, and Dipoles in Stratified Layers

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Overview

PyRAMIDS is a Python package for simulating electromagnetic radiation, magnetoelectric interactions, and dipole-based multiple scattering in stratified media. This manual briefly summarizes the code architecture and folder structure, lists external dependencies (with tested versions), and points to benchmarks and example scripts. Example cases include literature benchmarks, scripts to reproduce figures from the accompanying paper, and internal validation tests used for consistency checking across different definitions.

Installation and package dependencies

The PyRAMIDS source code is available via GitHub at:

`https://github.com/AMOLFResonantNanophotonics/PyRAMIDS`

The repository can be cloned using:

```
git clone https://github.com/AMOLFResonantNanophotonics/PyRAMIDS.git
```

PyRAMIDS is implemented in *Python* (v3.12.7) and distributed together with benchmark tests and usage examples. The computational core relies on *NumPy* (v1.26.4) and *SciPy* (v1.15.1), with extensive vectorization for numerical efficiency. Performance-critical routines are accelerated using *Numba* JIT compilation (v0.60.0), which can be enabled, disabled, or cached depending on user preference. Cached compilation is recommended for routine usage, as it combines compilation speed-up with minimal overhead during repeated runs. During development and debugging of core routines, disabling JIT is advisable. Visualization routines use *Matplotlib* (v3.9.2). A compatible Python environment can be created by installing the pinned dependencies, for example using `pip install package==version`. The versions listed above correspond to the development and testing environment used by the authors; PyRAMIDS should be generally compatible with earlier and more recent releases of these packages, although only the versions specified here have been formally validated.

Folder structure overview

The package is organized into the following top-level directories:

Library

Main computational framework implementing the mathematical formalism described in the Supplementary Material.

Benchmarks

Validation routines, including literature benchmarks and internal consistency tests.

Example

Typical usage workflows and scripts to reproduce figures from this paper.

UserSandbox

Workspace for user-developed scripts, including batch processing utilities to execute benchmarks and examples.

For quick orientation, a schematic of the repository layout is shown below.

PyRAMIDS: Directories at a glance

```
|-- Library/
| |-- Core/
| |-- Util/
| `-- Use/
|-- Benchmarks/
| |-- Literature/
| `-- Internal_Consistency/
|-- Example/
| |-- RoutineUse_Examples/
| `-- PaperReproduce_Examples/
`-- UserSandbox/
```

Library directory

The **Library** folder is divided into three subdirectories: **Core**, **Util**, and **Use**. This layered structure separates low-level mathematical formula implementations from user-facing interfaces.

Core

The **Core** directory contains low-level implementations written in a slab-centric coordinate system. These routines form the computational backbone of the package. Direct use is discouraged unless the user is extending the framework.

`Core_Smatrix.py`

Implements the S-matrix formalism for arbitrary stratified multilayer systems. The code constructs interface and propagation matrices and recursively builds the full stack response using the Redheffer star product for both s and p polarizations. Reflection and transmission coefficients in semi-infinite media are computed, along with internal field amplitudes.

`Core_ImGLDOS.py`

Computes electric, magnetic, and magnetoelectric LDOS using contour integration of the dyadic Green tensor. Adaptive integration is performed via `scipy.integrate.quad_vec`.

`Core_Radiationpattern.py`

Evaluates the far-field radiation pattern of an arbitrary magnetoelectric dipole $(p_x, p_y, p_z, m_x, m_y, m_z)$ embedded in a stratified medium using the asymptotic dyadic Green function. Fields are returned in the s-p basis for specified (θ, ϕ) angles. Integrated radiated power in each half-space is also provided.

`Core_Greenslab.py`

Computes generalized dyadic Green functions within a layered slab using the angular spectrum representation, assuming emitter and detector reside in the same layer.

Util

The **Util** directory provides helper functions that interface between user-level inputs and the Core routines.

`Util_argumentchecker.py`

Validates user input consistency such as dipole dimensionality, complex values, stack definitions, and many more. Reshapes arrays to match Core expectations.

`Util_argumentrewrapper.py`
: Converts user-centric coordinates (first interface at $z = 0$) into slab-centric coordinates for internal computations.

`Util_vectorpolarization.py`
: Provides visualization routines for far-field emission patterns and Stokes polarimetry, enabling Fourier-plane emission analysis analogous to experimental back-focal-plane imaging.

`Util_differential_evolution_custom_constraints.py`
: Provides a custom Differential Evolution optimizer with constraint handling. It is used for parameter optimization and inverse-design routines within the package.

Use

The **Use** directory contains the high-level user interface. These scripts operate in a global (user-centric) coordinate system and internally call Util and Core modules.

`Use_Planewaves.py`
: Computes reflectance, transmittance, absorptance, and layer-resolved absorption for s- and p-polarized plane waves incident at arbitrary angles. Includes field-distribution visualization routines.

`Use_LDOS.py`
: Computes electric, magnetic, and magnetoelectric LDOS for canonical and arbitrary dipole orientations at any position within the stack. Provides access to $k_{\parallel}$-resolved integrands for modal analysis.

`Use_Radiationpattern.py`
: Calculates angle-resolved far-field emission and radiated power. Radiative LDOS can be obtained via angular integration.

`Use_Green.py`
: Computes the full and scattered dyadic Green tensors in layered media (with source and detector in the same layer), including helper wrappers for user-centric coordinates.

`Use_Multiplescattering.py`
: Solves multiple-scattering problems for dipole scatterers in stratified media. The coupling matrix is constructed from inverse polarizabilities and Green-function interactions. The module computes induced dipole moments, extinction cross-sections (via work done), and scattering cross-sections (via far-field integration). Radiation-damping corrections consistent with the optical theorem are included.

Benchmarks directory

This folder contains two subdirectories:

Literature

Reproduces established results from Refs. [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17].

Internal_Consistency

Contains five internal diagnostic benchmark scripts:

`InternalBenchmark_LDOSvsRadpat.py`
: Validates that LDOS computed from Im G agrees with LDOS obtained from integrating the far-field radiation pattern, tested for electric, magnetic, and magnetoelectric dipoles, including off-axis positions and controlled phase shifts, in a non-absorbing, no-guided-mode test geometry.

`InternalBenchmark_GreenvsLDOSvsRadpat.py`
: Cross-compares 3 independent routes to LDOS-like quantities: (i) radiation-pattern integration, (ii) ImG-based LDOS, and (iii) values derived from the scattered dyadic Green tensor.

`InternalBenchmark_Green.py`
: Validates the dyadic Green-tensor implementation against (i) analytical free-space Green tensor, (ii) single-interface reflected contribution, and (iii) symmetry/sign relations under mirrored stack geometries.

`InternalBenchmark_multiplescattering.py`
: Verifies optical-theorem consistency for plane-wave-driven scattering by dipole arrays in layered structures, using a non-absorbing, no-guided-mode reference geometry.

`InternalBenchmark_PseudoChiralRotationRadPattern.py`
: Checks rotational consistency of pseudochiral dipole definitions via invariance of Stokes S_3 radiation maps in the upper and lower hemispheres.

These diagnostic checks ensure physical consistency and numerical robustness across the full framework.

Examples and UserSandbox directories

The **Example** directory provides scripts demonstrating standard workflows and reproducing figures from this chapter.

The **UserSandbox** directory serves as a customizable workspace for user-defined problems and includes batch-execution scripts for running all benchmark and example files.

References

[1] Ralf Ameling and Harald Giessen. Cavity plasmonics: large normal mode splitting of electric and magnetic particle plasmons induced by a photonic microcavity. *Nano Lett.*, 10(11):4394–4398, 2010.

[2] R. M. Amos and W. L. Barnes. Modification of the spontaneous emission rate of Eu^{3+} ions close to a thin metal mirror. *Phys. Rev. B*, 55:7249, 1997.

[3] William L Barnes, Simon AR Horsley, and Willem L Vos. Classical antennas, quantum emitters, and densities of optical states. *J. Opt.*, 22(7):073501, 2020.

[4] Lei Wei, Anatoly V Zayats, and Francisco J Rodríguez-Fortuño. Interferometric evanescent wave excitation of a nanoantenna for ultrasensitive displacement and phase metrology. *Phys. Rev. Lett.*, 121(19):193901, 2018.

[5] Yoel Fink, Joshua N Winn, Shanhui Fan, Chiping Chen, Jurgen Michel, John D Joannopoulos, and Edwin L Thomas. A dielectric omnidirectional reflector. *Science*, 282(5394):1679–1682, 1998.

[6] K. Joulain, R. Carminati, J. P. Mulet, and J. J. Greffet. Definition and measurement of the local density of electromagnetic states close to an interface. *Phys. Rev. B*, 68:245405, 2003.

[7] Andrej Kwadrin and A Femius Koenderink. Probing the electrodynamic local density of states with magnetoelectric point scatterers. *Phys. Rev. B*, 87(12):125123, 2013.

[8] Kwang Geol Lee, Xue Wen Chen, Hadi Eghlidi, Philipp Kukura, Robert Lettow, Alois Renn, Vahid Sandoghdar, and Stephan Götzinger. A planar dielectric antenna for directional single-photon emission and near-unity collection efficiency. *Nat. Photonics*, 5(3):166–169, 2011.

[9] Dongfang Li, Sinan Karaveli, Sébastien Cueff, Wenhao Li, and Rashid Zia. Probing the combined electromagnetic local density of optical states with quantum emitters supporting strong electric and magnetic transitions. *Phys. Rev. Lett.*, 121(22):227403, 2018.

[10] W Lukosz and RE Kunz. Light emission by magnetic and electric dipoles close to a plane dielectric interface. ii. radiation patterns of perpendicular oriented dipoles. *J. Opt. Soc. Am.*, 67(12):1615–1619, 1977.

[11] W Lukosz and RE Kunz. Light emission by magnetic and electric dipoles close to a plane dielectric interface. iii. radiation patterns of dipoles with arbitrary orientation. *J. Opt. Soc. Am.*, 69(11):1495–1503, 1979.

[12] Martin Neugebauer, Paweł Woźniak, Ankan Bag, Gerd Leuchs, and Peter Banzer. Polarization-controlled directional scattering for nanoscopic position sensing. *Nat. Commun.*, 7(1):11286, 2016.

[13] Lukas Novotny and Bert Hecht. *Principles of Nano-Optics*. Cambridge university press, 2012.

[14] Ivana Sersic, Christelle Tuambilangana, Tobias Kampfrath, and A Femius Koenderink. Magnetoelectric point scattering theory for metamaterial scatterers. *Phys. Rev. B*, 83(24):245102, 2011.

[15] Yichen Shen, Dexin Ye, Ivan Celanovic, Steven G Johnson, John D Joannopoulos, and Marin Soljačić. Optical broadband angular selectivity. *Science*, 343(6178):1499–1501, 2014.

[16] E Snoeks, A Lagendijk, and A Polman. Measuring and modifying the spontaneous emission rate of erbium near an interface. *Phys. Rev. Lett.*, 74(13):2459, 1995.

[17] H. P. Urbach and G. L. J. A. Rikken. Spontaneous emission from a dielectric slab. *Phys. Rev. A*, 57:3913, 1998.